Laws, the theory matrix, and experiments derived from laws

Memory families compared by proof

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A memory, abstractly

- **Study** an item: the memory's state changes.
- **Retrieve** with a probe: a probability distribution over stored items, or nothing.

Items are binary feature codes; similarity is the number of mismatching features.

Every law below is a property of this interface. It names no mechanism, but it does assume this representation.

Six laws

Certainty	An item just studied is retrieved by itself with probability 1.
Stability	Studying z leaves the chance that y retrieves itself unchanged.
Order-freedom	Swapping the last two study events leaves retrieval unchanged, for every probe.
Recency	Of two new items studied last, a probe equally similar to both retrieves the later more often.
Stored-only	Nothing is retrieved that was never studied.
Determinacy	Every retrieval has a single outcome.

Four families

- **Discrimination** (EPAM): a net of feature tests; learning grows a test where codes are confused.
- **Activation** (ACT-R declarative memory): each use adds activation that decays; partial matching; a failure weight. Decay 1, temperature 1: exactly rational.
- **Strength with a criterion** (after DOA): strengths only add; sample by strength, discounted by mismatch; state if above criterion.
- **Global matching** (MINERVA 2): every study stores a trace; a probe activates all traces by the cube of similarity; memory returns the *echo*, their weighted blend.

The matrix

	Certainty	Stability	Order-free	Recency	Stored-only	Determinate
Discrimination	**has**	**has**	lacks	lacks	**has**	**has**
Activation	**never**	**never**	**never**	**has**	**has**	**never**
Strength	lacks	lacks	**has**	**never**	**has**	lacks
Global matching	lacks	lacks	**has**	**never**	lacks	**has**

- **has** / **never**: a general proof that the law holds / fails at *every* parameter setting meeting stated conditions.
- lacks: a counterexample at *one* setting, computed exactly on the eight 3-bit codes.
- Every law holds for some family and fails for another; every family has its own row.

One setting, or every setting?

	at every setting	at one setting
the law holds	**has**	can hold
the law fails	**never**	lacks

- Only every-setting cells speak for the family, so only they can be refuted: by people lacking a law the family **has**, or having one it **never** has.
- A lacks cell refutes nothing: another setting may do what people do.
- Activation **never** has certainty, at any positive failure weight: retrieval can always fail.
- An order-free memory **never** has recency: either order of study leaves the same memory. Proved once for every memory; strength and global matching inherit it.
- The shape of SFT-GMA's critical tests: what an architecture *cannot* produce.

The laws were tested, and failed once

- The first four laws were chosen with three families in view.
- Global matching was added as a test. On those four laws, it had *exactly* the row of strength memory.
- So the laws missed something. What is distinctive about global matching? Its output is a blend.
- Study 110, 101, 011; probe with 111 (their prototype, never studied): MINERVA returns 111.
- Hence the stored-only law, which separates them.

Rule: a law enters only with a witness on each side.

Against the evidence

	Certainty	Stability	Order-free	Recency	Stored-only	Determinate
People (judged)	*lacks*	*mixed*	*lacks*	*has*	*mixed*	*mixed*

- A judgment, not a theorem, with sources, kept apart from the proofs.
- Each entry states its **bridge**: how a measure (a report) is taken to show a law (about retrieval).

Family	refuted on (an every-setting cell)	open on (one setting only)
Discrimination	certainty	recency
Activation	—	—
Strength	order-freedom, recency	—
Global matching	order-freedom, recency	—

One experiment, four memories: study, then test

Study 110, 101, 011. Probe each; then the lures 111 (prototype) and 000.

Memory	curve by position	111 called old	111 retrieves itself	000 called old
EPAM	$1, 1, 1$	1	0	1
ACT-R	$\frac{40}{97}, \frac{15}{28}, \frac{120}{157}$	$\frac{55}{61}$	0	$\frac{55}{67}$
Strength	$\frac{8}{17}, \frac{8}{17}, \frac{8}{17}$	1	0	1
MINERVA	$1, 1, 1$	0	1	0

- Derived from laws: EPAM's flat curve (certainty + stability $\Rightarrow$ flat at 1, for any memory).
- Computed: everything else. Recency does not imply a rising self-retrieval curve.
- "Called old" is a report; stored-only is about retrieval. EPAM calls the prototype old by retrieving a studied item.

Reversing the workflow

Usually: choose an experiment, compute each theory's predictions.

Instead:

1. find two laws that disagree;
2. derive a design on which they make incompatible predictions;
3. prove the separation, for every memory with either law;
4. state the response model that connects it to what participants say.

Recency against order-freedom

Design. Two new items a, b; a probe x equally similar to both. Condition AB studies a then b; condition BA studies b then a; both probe with x.

$$D = P(\text{retrieve } b) - P(\text{retrieve } a), \qquad \Delta = D_{AB} - D_{BA}.$$

Theorems, for any memory:

- order-freedom $\Rightarrow \Delta = 0$;
- recency $\Rightarrow \Delta > 0$;
- so no memory has both laws, wherever such a design exists.

From retrieval to report

Forced choice: report A if a is retrieved, B if b is; otherwise guess B with probability γ.

$$\Delta_{\text{report}} = \Delta + (2\gamma - 1)\,(o_{AB} - o_{BA}),$$

where o is the chance of retrieving neither item.

- Order-freedom: $\Delta_{\text{report}} = 0$, for every γ.
- Recency: $\Delta_{\text{report}} > 0$ with unbiased guessing, or when "neither" is equally likely in both orders.
- A biased guesser can mask or reverse the recency prediction.

The same design, four memories

Memory	Δ, no prior item	Δ, after 000	why
ACT-R	$10/17$		recency, so $\Delta > 0$
Strength	0	0	order-free
MINERVA	0	0	order-free
EPAM	0	-2	neither law: computed

- ACT-R's reported effect is $10/17$ for every guessing bias: both orders leave the same chance of "neither".
- EPAM shows the most extreme *primacy* possible. After 000, whichever item is learned first is discriminated from it, and the probe always retrieves that one. Its order effects come from the shape of the net, not from time.

Lessons, and next

- The durable results are conditional: laws $\Rightarrow$ predictions, families $\Rightarrow$ laws.
- The empirical row will change; when it does, the matrix says what is refuted.
- **Observational semantics** is the main gap: decision and report models, so retrieval laws predict responses.
- **Parameter regions** of the "lacks" cells; **graded laws**; **more families** as tests (REM, connectionist memory, ACT-R with blending).
- **Beyond memory**: strategy choice, attention, production systems.